Review: Research Design & Advanced Topics

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Applied Quantitative Methods II
MA in Social Sciences, Spring 2026

Today's second half

- The course has been about **estimation**: how to fit models
- But fitting a model is rarely the hard part of empirical work
- Today: what you actually need to answer a research question
  - → Research design and identification
  - → Causal inference methods beyond FE/DiD
  - → Defending your results: robustness and validity
  - → Common pitfalls when applying all of this
- Plus a pointer to two advanced topics (not covered today)

This lecture is deliberately different in tone from the rest of the course. We have spent the semester learning how to estimate a wide range of models. What I want to do today is to step back and make explicit the framing that sits behind all of that: what a regression actually gives you, what it does not, and what additional design choices make the difference between a credible empirical claim and a suggestive correlation. The goal is not to teach new techniques in the last hour of the course, but to give students a mental map they can carry into their final essay and into whatever they do next. The appendix at the end has pointers for two topics — multilevel models and text as data — that I will not cover in class but that students may want to explore on their own.

Roadmap

From estimation to identification

Causal inference toolkit

Defending a causal claim

Choosing the right method & pitfalls

What you have learned

- Estimation tools for many kinds of outcomes
  - → OLS, logit, count / ordinal / duration models
- Designs that exploit data structure
  - → Fixed effects, two-way FE, DiD (including staggered)
  - → Spatial models
- How to interpret and present results
  - → Predicted values, marginal effects, coefficient plots
- Plus computing practices to make all of this reproducible

Quick inventory of what the course has covered. The common thread is: given a research question and data, how do you go from raw numbers to a defensible estimate and a readable figure. That is real and useful work — but it is only one piece of what empirical research actually requires. The next slides shift the frame.

Two different questions

Estimation: given a model, what are the parameters?

Identification: do those parameters mean what I think they mean?

Your R output answers the first.

Only your research design can answer the second.

This is the central distinction. Estimation is a statistical problem: given a specification, compute the best-fit coefficients. R will always give you a number. Identification is a research-design problem: does the coefficient you just estimated correspond to the causal (or descriptive) quantity you actually want to learn about? The reason identification is hard is that the same regression output is consistent with many different underlying data-generating processes, and you cannot tell them apart from the estimation alone. You need theory and design assumptions.

The fundamental problem of causal inference

- A **causal effect** is the difference between what *did* happen and what *would have* happened otherwise
- For unit i with treatment D:

$$\tau_i = Y_i(1) - Y_i(0)$$

- We only ever observe **one** of these two worlds
  - → If i was treated, $Y_i(1)$ is observed and $Y_i(0)$ is missing
  - → If i was not treated, the opposite
- Causal inference = principled way to estimate the missing counterfactual

The potential outcomes framework in one slide. Emphasize that this is a missing data problem in disguise: for any given unit, half of the information we would need to compute a causal effect is unobservable. Everything else — randomization, fixed effects, DiD, IV, RDD, matching — is machinery for building a credible estimate of that missing counterfactual. If the counterfactual is not credible, the machinery does not help.

What we can estimate on average

- We can't recover τ_i for individuals
- But we can target averages:
  - → **ATE** — average over everyone: $E[Y(1) - Y(0)]$
  - → **ATT** — average over the treated: $E[Y(1) - Y(0) \mid D = 1]$
  - → **LATE** — average over those who *respond* to a treatment (IV, RDD)
- Different methods estimate different quantities
  - → When you compare two papers: are they even targeting the same thing?

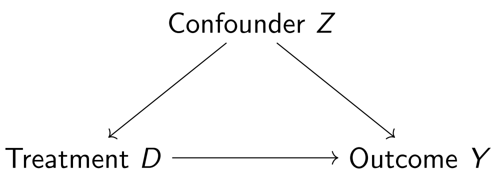
A useful sanity-check when reading applied papers: which of these averages is the estimator targeting? A randomized experiment gives you the ATE over the experimental sample. A DiD gives you the ATT on the treated units. An IV or RDD gives you a LATE, a local effect for those whose treatment status is actually driven by the instrument or the cutoff. These are different quantities. Two papers can report conflicting effects on the same phenomenon simply because they are estimating different averages over different subpopulations.

Causal models and DAGs

- To approximate the counterfactual we need a model of *why* units are treated in the first place
- Directed Acyclic Graphs (DAGs): nodes are variables, arrows are direct causal effects
- Three structures you must recognize:
  - → **Confounder** $D \leftarrow Z \rightarrow Y$
  - → **Mediator** $D \rightarrow M \rightarrow Y$
  - → **Collider** $D \rightarrow C \leftarrow Y$
- What you control for — and what you do **not** — depends on which structure you are looking at

DAGs are the most compact way to talk about what a regression is really doing when you add a control variable. The three structures on this slide cover most of the cases you will see. The non-obvious rule: you want to control for confounders, you should not control for mediators (that blocks the effect you are trying to measure), and you should absolutely not control for colliders (that opens a spurious path).

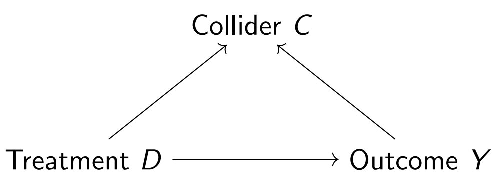
Confounding: the standard case



- Z affects both D and Y: a back-door path
- Un-adjusted comparison of treated vs. untreated mixes two effects
- **Solution:** condition on Z (or use a design that blocks the back door)

Ice-cream sales and drownings; pirate numbers and global temperature. These are the textbook examples, but confounding is also the implicit story behind every "we add controls for X, Y, Z" paragraph in an empirical paper. Every control you add should correspond to a node on a DAG that your theory justifies. Adding controls because they change the coefficient is how you generate p-hacked results.

Colliders: do not control!



- C is caused *by* both D and Y
- Conditioning on C creates a spurious association between D and Y
- Classic: "smart people lack social skills" (if you condition on being hired)

The counterintuitive result in causal inference: some variables make the bias worse when you add them as controls. A collider is any variable that is downstream of both your treatment and your outcome. Conditioning on it opens a non-causal path between the two. The standard illustration: intelligence and social skills may be uncorrelated in the population, but if you restrict attention to people hired at a competitive company (which selects on both), you will find a spurious negative correlation.

Post-treatment bias

- Special case of collider bias that is very easy to miss
- A **post-treatment** variable is one that is itself affected by the treatment
- Controlling for it:
  - → blocks part of the mechanism (if it is a mediator), **or**
  - → opens a collider path (if it is also affected by other unobservables)
- Rule of thumb: if in doubt about whether a variable is pre- or post-treatment, leave it out

Post-treatment bias is the single most common avoidable mistake in applied work. Typical pattern: researcher estimates the effect of a treatment, worries that results might be confounded, throws in every variable available including some that were measured after the treatment. Those post-treatment variables either sit on the causal pathway the researcher is trying to measure (so controlling for them attenuates the effect mechanically) or open collider paths (so controlling for them introduces bias). Montgomery, Nyhan and Torres 2018 AJPS is the reference; keep it in mind when you are writing your final essay.

So what does your regression assume?

Every regression coefficient you report implicitly assumes:

1. You have **closed all back doors** (no omitted confounders)
2. You have **not opened new ones** (no controlling for colliders or mediators)
3. The functional form is roughly right (linearity, additivity)

(1) and (2) are **design** assumptions, not something you can test.

Wrap up the first section. The assumptions that matter most are the ones that cannot be checked from the data alone — they come from your theory and your argument about the data-generating process. This is why a paper's identification strategy is more important than its R-squared.

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What you already have

Two of the four big causal-inference designs are already in your toolkit:

- **Fixed effects** — control for all time-invariant unobservables at some level (unit, region, year...)
- **Difference-in-differences** — use a never-treated comparison group plus pre/post variation; identifies the ATT under parallel trends

Two more that you should at least recognize:

- **Regression discontinuity design (RDD)**
- **Instrumental variables (IV)**

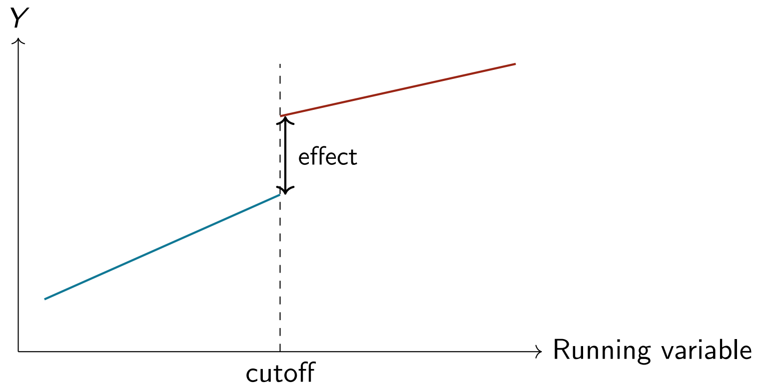
Recap what we have, flag what we still need to cover at a high level. I do not expect students to leave this lecture able to run an RDD or an IV regression — I want them to recognize these designs when they encounter them, understand the intuition, and know when a problem calls for them.

Regression discontinuity design

- **Setup:** treatment assignment depends on whether a *running variable* crosses a cutoff
  - → Incumbency advantage: winners of close elections
  - → Scholarships: eligibility based on test score cutoff
  - → Conscription: draft lottery numbers below a threshold
- **Idea:** units just above and just below the cutoff should be similar in everything *except* the treatment
- Estimates a LATE: effect for units near the threshold

RDD has become one of the most popular designs in political science over the last decade, largely because close-election data is readily available and the identification assumption is transparent. Important caveat students should know: an RDD estimate only tells you about units near the cutoff. The effect of a scholarship for marginal admits may look very different from the effect for a student who was a clear admit.

RDD visually



- The jump at the cutoff is the treatment effect
- The trend on each side controls for whatever else varies smoothly

This picture is the whole argument. We believe any confounders vary smoothly as the running variable moves. At the cutoff, treatment changes discretely while everything else stays continuous. So a discontinuity in the outcome at that point can only be caused by the treatment. The design can fail if people can precisely manipulate their position around the cutoff, or if the cutoff triggers other changes besides the treatment of interest.

Instrumental variables

$$\begin{tikzcd} & & U \arrow[dl, dashed] \arrow[dr, dashed] & \\ \text{Instrument } Z \arrow[r] & \text{Treatment } D \arrow[rr] & & \text{Outcome } Y \end{tikzcd}$$

- Find a variable Z that moves D but has no other path to Y
- Use only the variation in D explained by Z
- Two assumptions:
  - → **Relevance**: Z actually affects D (testable)
  - → **Exclusion restriction**: Z affects Y only through D (not testable)

The IV design. Classic examples: rainfall as an instrument for protest intensity (rainfall shifts who shows up to a protest, but should not affect whatever political outcome you are studying except through its effect on protests). The canonical problem with IV in practice is the exclusion restriction: you have to tell a story about why the instrument could not plausibly affect the outcome by any route other than through the treatment. That story is often contested.

The four design templates

Design	**Exogenous variation**	**Key assumption**
FE	Within-unit differences	No time-varying confounders
DiD	Treated vs. comparison, pre/post	Parallel trends
RDD	Jump at cutoff	No manipulation at cutoff
IV	Instrument → treatment	Exclusion restriction

- You do not pick a design, *the data* pick one for you
- What source of exogenous variation does the world hand you?

The design-based approach to causal inference, in one slide. A good applied paper starts from a research question and a dataset and asks: what source of variation in this setting can I credibly treat as exogenous? The answer to that question determines the method. It is very rarely the case that the researcher picks the method first and then looks for a matching application.

Matching: complement, not substitute

- Not a design in the same sense — but a useful preprocessing step
- **Idea:** pair each treated unit with one or more untreated units that look similar on observed covariates
- Compare only within matched pairs
- What it fixes: imbalance in observed covariates and reliance on linear-in-parameters extrapolation
- What it does **not** fix: omitted confounders, selection on unobservables
- Often used *with* regression or DiD, not instead of

Matching is sometimes presented as a rival to regression; it is better understood as a complement. Both methods deal with the same problem — conditioning on observed covariates — but matching makes the comparisons more transparent and less dependent on functional-form assumptions. If the two agree, you are on firmer ground. If they disagree, something about your specification is doing a lot of the work.

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Your main result is the beginning

- You have a point estimate, a standard error, a figure
- A skeptical reader should be asking:
  - → What if you had specified it differently?
  - → What if your identifying assumption is wrong?
  - → What if you are measuring the wrong thing?
  - → Would the result generalize elsewhere?
- Your job is to answer those questions **before they do**

Framing for the third section. A published empirical paper is not just the main result; it is the main result plus a set of complementary analyses that demonstrate the result is not an artifact of specification, assumption, measurement, or sample. If students take one piece of practical advice from this lecture, it should be: plan those complementary analyses before you run the main model, not afterwards as damage control.

Robustness: alternative specifications

- Change one thing at a time, show it does not matter much
- Measurement of Y: alternative indicators, different scales
- Measurement of D: dichotomize vs. continuous, source
- Sample: drop influential units, drop specific sub-groups
- Model: add/remove controls, change FE structure, change SEs
- Functional form: log, quadratic, non-parametric
- Display as a coefficient plot across specifications, not a 10-column table no one reads

When you change a measurement choice or a modeling choice, the coefficient should move, but the substantive story should not. The goal is not to show that the point estimate is identical across specifications (it won't be); it is to show that the textitconclusion is robust. The visualization matters — a specification curve or a coefficient plot across many specifications is much more convincing than a kitchen-sink regression table.

Placebo tests

- Look for an effect *where there should be none*
- If you find it, something is wrong with your design
- Three flavors (Eggers, Tuñón & Dafoe 2023):
  - → **Placebo outcome**: different Y that the treatment should not affect
  - → **Placebo treatment**: a fake D that should have no effect
  - → **Placebo population**: a subgroup not exposed to D

Placebos are the most intuitive tool for checking an identification assumption. If you claim that your effect is driven by a specific mechanism, there should be outcomes, treatments, or populations where that mechanism cannot plausibly operate. If you find the same-sized effect there too, that is evidence your identification is picking up something other than what you think. A classic example is the Peisakhin & Rozenas study on Russian TV: they show that sports-channel signal strength does not predict voting, which builds confidence that the news-channel result is about content and not about something generic like TV reception quality.

Placebo tests in DiD

- Most important placebo in a DiD setup: **pre-trends**
- Before the treatment, the two groups should move in parallel
- If pre-treatment "leads" are significant, parallel trends is suspect
- Event-study plots show this visually:
  - → Coefficients on event-time dummies
  - → Pre-period coefficients should be flat and near zero
  - → Post-period coefficients show the dynamic effect

This ties back to Session 6. Event-study plots are the workhorse visualization for modern DiD work. The pre-treatment portion is itself a placebo test: if the treatment has not happened yet, there should be no treatment effect. A slope in the pre-period is evidence that your comparison group was not a good counterfactual to begin with.

Additional implications

- If your theory is correct, what *else* should be true in the data?
- **Mechanism checks**: intermediate variables your theory predicts
- **Heterogeneity**: groups where the effect should be larger or smaller
- **Rule-out tests**: observable patterns that competing explanations would imply but yours would not
- A paper that only shows the main effect is weaker than one that shows the main effect **plus** two or three such auxiliary findings

Strong empirical papers do not just test their main hypothesis; they derive and test auxiliary implications of their theory. This is where the research-design framing pays off: a well-specified theory will imply patterns of heterogeneity (the effect should be larger among group X), mechanism predictions (intermediate variable M should move with the treatment), and exclusion patterns (if the competing explanation were right, we would see Y, but we see Z instead). This is what turns an empirical paper from "I ran a regression" into "I used the data to discriminate between competing accounts."

Measurement and construct validity

- Are you measuring what you think you are measuring?
- **Treatment**: when you say $D = 1$, what is actually being delivered?
  - → Bundled treatments (media exposure = content + format + novelty...)
- **Outcome**: does the indicator really capture the concept?
  - → Polarization = partisan thermometer gap? policy-view distance? affective feelings?
- Identification can be perfect and the claim still be wrong if the measurement doesn't line up with the theory

Measurement is underrated. A perfectly identified estimate of something that is not what you wanted to measure is not progress. Ask, for any paper you read: what would have to be true for the reported indicator to really capture the concept the author talks about in the introduction? Often the answer is "a lot" and that creates space for disagreement even when the design is clean.

External validity

- Under what *conditions* does the effect hold?
- Four dimensions (Egami & Hartman 2023):
  - → **Populations**: units beyond your sample
  - → **Treatments**: variants of the intervention
  - → **Outcomes**: related but different Ys
  - → **Contexts**: other settings, places, times
- A clean estimate in one sample does not automatically generalize
- **Temporal validity** (Munger 2023): does it still hold 10 years from now?

External validity is the question most often brushed aside with a single paragraph in the conclusion. Egami and Hartman offer a usable framework: rather than treating external validity as a yes/no question, decompose it into specific dimensions and ask which ones plausibly transfer and which do not. The temporal dimension is particularly salient in fast-moving domains like social media research, where the platform you studied five years ago may not exist in the same form today.

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Starting point: your question type

Goal	Tools
Describe a pattern	Summary stats, visualization, simple regression
Predict Y from X	Regression, ML (cross-validated)
Explain Y causally	Design-based methods: FE, DiD, RDD, IV, experiments
Generalize a mechanism	Theory + replication across contexts

- Most confusion in applied work comes from mixing these up
  - → e.g. treating predictive coefficients as causal

If you take one thing from this slide: be explicit about what kind of question you are trying to answer, because the standards of evidence differ. A predictive model can be judged on out-of-sample performance and does not need causal identification. A causal claim needs an identification strategy and out-of-sample performance is secondary. A descriptive paper needs neither but requires careful measurement and framing. Papers that slide between these modes are weaker than papers that commit to one.

Starting point: your outcome

Outcome type	Method
Continuous	OLS (always a good first pass)
Binary	Logit, probit (or LPM for interpretability)
Count (non-negative integer)	Poisson, negative binomial
Ordered categories	Ordered logit / probit
Time-to-event	Cox, Kaplan–Meier, parametric survival
Panel data	+ fixed effects, + clustered SEs
Spatial data	+ spatial weights, + Moran's I

- This is a cheat-sheet for your final essay — keep it

The cheat-sheet version of the whole course. This is not a substitute for understanding what each model does; it is a decision aid for when you are staring at a dataset and trying to pick a starting point. For most social science questions, the answer is "OLS first, then something more specialized if the defaults are clearly wrong."

Pitfall 1: p-hacking and forking paths

- Garden of forking paths (Gelman & Loken 2013)
- Every analysis involves many small choices:
  - → Which controls to include
  - → How to code the treatment
  - → Which subsample to focus on
  - → How to handle outliers
- If any of those choices were made *after looking at the results*, your p-values are too small
- Mitigations: pre-analysis plans, specification curves, honest reporting of all tried specifications

P-hacking is not always deliberate. Most of it happens because researchers legitimately try several specifications, find one that "works" and report that one. The statistical problem is that if you search across 20 specifications and report the one that gives p ¡ 0.05, you have inflated your Type I error rate enormously. The honest solutions are either to pre-register your analysis (commit before seeing the data) or to show the full distribution of specifications you considered.

Pitfall 2: bad controls

- "More controls" is not always better
- **Do not control for**:
  - → Mediators (blocks the effect you want to measure)
  - → Colliders (opens a spurious path)
  - → Post-treatment variables (same problem, often both)
  - → Proxies for the treatment (collinearity)
- Every control needs a justification from your DAG
- Cinelli, Forney & Pearl (2024) "A crash course in good and bad controls"

Classic failure mode in applied work: the reviewer asks for robustness to additional controls, the researcher adds every available variable, and the estimate swings around. Often what has happened is that some of those added controls are post-treatment or collider variables, and adding them has introduced bias rather than reducing it. The right response to "add controls" is not automatic compliance; it is a DAG-based argument for which controls are needed and why.

Pitfall 3: ignoring dependence

- Standard errors assume observations are independent
- In social science data they usually are not:
  - → Observations nested within countries, schools, firms
  - → Same unit observed over time (panel)
  - → Spatial proximity
  - → Network connections
- Consequence: default SEs can be *much* too small
- Fixes: cluster-robust SEs, Conley SEs (spatial), network-adjusted SEs

Not glamorous, but very important: a paper with the correct identification but wrong standard errors can look convincing and be wrong. Cluster at the level of treatment assignment (Abadie, Athey, Imbens and Wooldridge 2023 is the modern reference). For spatial data, Conley standard errors. For network data, there are bespoke approaches. The main message: always ask at what level the dependence in your data might operate, and adjust accordingly.

What's next (not in this class)

- **Multilevel / hierarchical models** (Gelman & Hill)
  - → Natural framework when data are nested in groups
- **Bayesian inference**
  - → Priors, posteriors, and what MCMC actually does
  - → McElreath, *Statistical Rethinking*
- **Machine learning for social science**
  - → Prediction-first tools: lasso, random forests, boosting
  - → Causal ML (honest trees, double/debiased ML)
- **Text as data**
  - → From bag-of-words to LLMs in social science

Pointers rather than content. I will put a bit more on multilevel models and text as data in the appendix of these slides; the others are entire courses on their own, but the references listed here are the standard starting points. The McElreath book in particular is the most approachable introduction to Bayesian thinking aimed at social scientists.

A checklist for your final essay

- What is the **question** (descriptive, predictive, causal)?
- What is the **unit** of analysis and where does **variation** come from?
- What is the **DAG** behind the relationship?
- What method matches the question and the variation?
- What are the **identifying assumptions**, and how plausible are they?
- What **alternative explanations** does the design *not* rule out?
- What **robustness / placebo** checks can you run?
- How do you **measure** your key concepts, and what's lost in that translation?

The checklist students should keep in front of them as they write the essay. None of these questions is solved by R code. All of them affect whether the final paper is any good.

Good luck with the essay.

Appendix follows: multilevel models and text as data.
Self-study material, not covered in class.

Roadmap

Appendix: Multilevel models

Appendix: Text as data

Hierarchical data

- Very common structure in social science:
  - → Students within schools within districts
  - → Voters within precincts within countries
  - → Respondents within country-years
- Two facts:
  - → Units within the same group are more alike than units in different groups
  - → Groups vary in how they respond to predictors
- Ordinary OLS ignores both

Multilevel models (also called hierarchical or mixed-effects models) are built for this situation. The canonical textbook is Gelman and Hill (2007), or Gelman, Hill and Vehtari (2021) for the newer edition. The framework generalizes the fixed-effects ideas you have already seen and places them in a Bayesian (or hierarchical-likelihood) context.

Three ways to handle groups

- **Complete pooling**: ignore groups entirely, one intercept
  - → Wastes the group-level information
- **No pooling**: separate intercept for each group (fixed effects)
  - → Fits each group exactly — but noisily for small groups
- **Partial pooling**: each group gets its own intercept, but shrunk toward a global mean
  - → Best of both worlds — this is what multilevel models do

The key insight of multilevel modeling is partial pooling. If you have 50 students in one school, you can trust the school's mean; if you have 3 students in another, the school-specific mean is unreliable and you should borrow information from the overall distribution. Multilevel models do this automatically via the estimated group-level variance, shrinking small-sample groups more strongly toward the global mean.

Random intercepts

$$y_{ij} = \alpha_j + \beta x_{ij} + \varepsilon_{ij}, \quad \alpha_j \sim \mathcal{N}(\mu_\alpha, \sigma_\alpha^2)$$

- Each group j has its own intercept α_j
- But the α_j's are drawn from a common distribution
- Estimate: one global mean + one variance, not 200 separate intercepts
- Versus classic FE: FE *estimates* each α_j; random intercepts *model* them

The formal difference between fixed effects and random intercepts is whether you treat the group intercepts as fixed parameters to estimate (FE) or as draws from a distribution whose parameters you estimate (random intercepts). The practical difference is the shrinkage: random intercepts borrow strength across groups, which matters a lot when some groups have few observations. The trade-off: random intercepts assume the group intercepts are uncorrelated with the covariates. If that is violated (which is often the case in causal settings), fixed effects are the safer choice. This is the core of the Hausman test discussion from Session 6.

Random slopes

$$y_{ij} = \alpha_j + \beta_j x_{ij} + \varepsilon_{ij}$$

- Now *both* intercept and slope vary by group
- Effect of x differs across schools / countries / firms
- Heterogeneity is explicitly modeled, not averaged away
- Useful when you expect the effect itself to depend on context

Random slopes are what you want when the effect of the predictor itself varies across groups. A common example: the effect of a school-level policy on student outcomes may differ from one district to another. With a random slope you estimate the distribution of district-specific effects, not just the average. This is also a natural way to test for effect heterogeneity.

Fitting multilevel models in R

- Frequentist / REML: `lme4`
  - → `lmer(y ~ x + (1 | group), data = d)` — random intercept
  - → `lmer(y ~ x + (1 + x | group), data = d)` — random slope
- Bayesian: `brms` or `rstanarm`
  - → Same formula syntax, full posterior draws
- Visualization: `sjPlot::plot_model()`, `broom.mixed`

Quick practical pointers. If a student wants to experiment, start with
texttttlme4::lmer on a simple random-intercept model. The syntax is similar to
texttttlm with the addition of the
texttt(1 — group) term. Once the basics are comfortable, moving to
texttttbrms gives you a Bayesian version with priors, posterior visualization, and easy extension to non-Gaussian outcomes.

Multilevel vs. fixed effects

	Multilevel	Fixed effects
Goal	Describe variation across groups	Remove group-level confounding
Assumes	Group effects uncorrelated with x	Nothing about that
Small groups	Handles well (shrinkage)	Noisy
Causal setting	Use if assumption plausible	Safer default

- If your goal is descriptive and you have many small groups, multilevel is natural
- If your goal is causal and you worry about group-level confounding, FE is safer

The practical rule of thumb. The two approaches answer slightly different questions. Multilevel is a descriptive and predictive tool by default; fixed effects is a causal identification tool. They can be combined (fixed effects at one level, random effects at another), and in practice many modern applied papers do exactly that.

Multilevel: where to go next

- Gelman & Hill (2007) *Data Analysis Using Regression and Multilevel/Hierarchical Models*
- Gelman, Hill & Vehtari (2021) *Regression and Other Stories*, ch. 21–22
- McElreath *Statistical Rethinking*, ch. 13–14
- Bolker's `GLMM FAQ`: bbolker.github.io/mixedmodels-misc/glmmFAQ.html

The standard reading list. Gelman and Hill is the classic; the newer textitRegression and Other Stories is more accessible and covers the same material in a modern style. McElreath is excellent if you want the Bayesian perspective from the start.

Roadmap

Appendix: Multilevel models

Appendix: Text as data

Why text?

- Political science has always worked with text
  → Speeches, manifestos, news, laws, tweets, transcripts
- Historically: close reading and hand-coding
- Now: the volume makes automation essential
- A quantitative approach: turn documents into features, treat them like any other variable

Text is the dominant form of political data: everything from a congressional speech to a Twitter post to a newspaper headline is a text. The explosion of computational methods for processing text has made large-scale analysis feasible in a way it simply was not 15 years ago. The goal of this short appendix is to give a map of the main approaches, not to teach any of them in detail.

Preprocessing

- Tokenization: split text into words or sub-words
- Lowercasing
- Remove punctuation, numbers, stopwords ("the", "and")
- Stemming / lemmatization: "running", "ran", "runs" → "run"
- Every decision changes the downstream analysis
- Document these choices the way you would any other transformation

Preprocessing looks mechanical but it's actually a set of researcher-degrees-of-freedom. Whether you stem, whether you remove stop-words, whether you keep hashtags, whether you treat bigrams as separate tokens — all of these affect your results. They should be reported and justified. The
textttquanteda package in R handles these steps in a reproducible pipeline.

Bag of words

- Simplest representation: count how often each word appears
- Build a **document-term matrix (DTM)**
  - → Rows: documents
  - → Columns: unique words
  - → Cells: counts (or TF-IDF weights)
- Ignores word order and grammar
- Surprisingly effective for many tasks

The bag of words is the workhorse of classical text analysis. Despite throwing away syntax, it captures enough about topic and vocabulary to do a lot of useful work — classification, clustering, dictionary matching, topic modeling. When people criticize these methods for being "too simple" the honest response is that they are a strong baseline and often outperform more complex models on small datasets.

Dictionary methods

- Define a list of words associated with a concept
  - → Sentiment: “good”, ”great”, “bad”...
  - → Populism: “elite”, ”people”, “corrupt”...
- Count occurrences per document, maybe normalize
- Pro: simple, transparent, reproducible
- Con: brittle to context, sarcasm, domain shift
- Classic in political science — see Laver, Benoit & Garry (2003) Wordscores; LIWC

Dictionary methods are still widely used in political science because they are transparent and auditable. You can show exactly why a document was scored the way it was. The failure mode is obvious: the same word can mean very different things in different contexts, and hand-curated dictionaries often do not travel well across domains.

Topic models

- Unsupervised: discover themes in a corpus
- Latent Dirichlet Allocation (LDA) and descendants
  - → Each document is a mixture of topics
  - → Each topic is a distribution over words
- Structural Topic Model (STM, Roberts et al.)
  - → Includes document-level covariates (author, year, party)
- R packages: `topicmodels`, `stm`

Topic models are useful for exploratory analysis of large text corpora — getting a sense of what is in there without reading every document. The political-science-specific contribution here is Roberts, Stewart and Tingley’s Structural Topic Model, which lets you include covariates at the document level. This is the go-to tool if you want to ask questions like “do Republican senators talk about the economy differently from Democrats?”

Supervised classification

- You have labels for some documents; predict labels for the rest
- Examples:
  - → Is this speech about foreign policy? (yes / no)
  - → Is this tweet hate speech? (multi-class)
- Pipeline: DTM or embeddings → classifier (logistic, SVM, random forest)
- Evaluation: held-out set, cross-validation, F1/precision/recall

Supervised methods are where text-as-data overlaps most with traditional machine learning. The reason to do this rather than hand-code everything is scale: a trained classifier can label a million documents overnight. The reason to be careful is that the labels you train on determine what the classifier learns, and biased training data produces biased classifiers.

Embeddings and LLMs

- Represent words (or documents) as dense vectors that encode meaning
- word2vec, GloVe, sentence-BERT...
- Captures word *similarity* ("king" - "man" + "woman" $\approx$ "queen")
- Large language models (GPT, Llama, etc.)
  - → Zero-shot classification, extraction, annotation
  - → Replacing hand-coders for many tasks
  - → But: black box, bias, replication concerns

The current frontier. Embeddings have been around for about a decade and are now baseline tools; LLMs are newer and are actively changing what counts as a reasonable workflow for many text tasks. The reproducibility issues are real: if you use a closed commercial API, the model may be updated or retired and your results may not be replicable. Open-weights models are preferable when this matters.

Text as data: where to go next

- Grimmer, Roberts & Stewart (2022) *Text as Data*. Princeton UP
  - → The standard reference for social scientists
- R packages worth knowing:
  - → `quanteda` — full preprocessing & analysis pipeline
  - → `stm` — structural topic models
  - → `text` — interface to transformer embeddings
- Course: Pablo Barberá's text-as-data materials (online)

The Grimmer, Roberts and Stewart book is the starting point — it is aimed at social scientists, uses political-science examples, and covers both classical and modern methods. Pablo Barber
'a teaches a well-known text-as-data course, and his teaching materials are available online for anyone who wants to follow along.

Thanks, and good luck.